Wenjie Luo

Agent Application Engineer | Graduating Jan 2027

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github.com/LUOaini1213 | luoaini1213.github.io (demos, source and evaluation reports)

PROFILE

- Build Python and TypeScript agent applications with deterministic tools and human approval; turn business workflows into APIs, state transitions, execution traces and automated acceptance tests.
- Also work on GPU computing and inference-performance attribution, backed by single-variable controlled comparisons and reproducible runs.

EDUCATION

National University of Singapore | MSc Civil Engineering (Transportation) Aug 2025 - Jan 2027
Coursework incl. ML Foundations (CEE5901QA), Advanced AI for CEE (CEE5901QB), intelligent transport and simulation.

Chang'an University | BEng Transportation Engineering Sep 2021 - Jun 2025
GPA 3.16/5.0; approximately top 15% of the cohort.

PROJECTS

Civil Buddy | Agent Runtime and Tools (Rust workbench + MCP, Python, FastAPI) Jul 2026 - Present
- Built session scheduling, a tool engine, policy checks and recovery for 66 role workflows; high-risk writes require human confirmation; the policy engine covers privilege denial, tool-failure retry and degradation, and cost-limit circuit breaking.
- Linked packing, confirmation and visualization; 128/128 evaluation runs completed (57 judged not to fit and handed back to a person). Shadow tests compare predefined steps with tool calling, with a keyless policy fallback.

Glass Box | Agent Observability Middleware (TypeScript / Fastify) Aug 2026 - Present
- Built observability on the official starter: parent-child span waterfalls show tools, tokens, estimated costs and retries, with trace queries and OTLP/JSON export; the local durable queue passes 503, timeout and killed-process recovery tests.
- Secret-exfiltration policy blocks at a pre-run gate and during the run; Node 22/24 CI runs type checks, tests, a build, replay smoke (complete, fail, deny) and OTLP recovery smoke.

EDA Copilot | Traceable Engineering Q&A (RAG) Sep 2026
- Ran OpenROAD's nangate45/gcd example from RTL to GDSII; indexed reports, logs and docs into 1,215 chunks with source line references; built BM25, TF-IDF and hybrid retrieval.
- On self-authored offline tests, BM25 and hybrid both reached Hit@5 of 22/23; guardrails stopped all 12 should-ask cases with correct reason codes, versus none when disabled.

Counterask | Conversational WebMCP Storefront Sep 2026
- Serverless single-page store with 11 imperative tools and a declarative checkout the agent can fill but only the user can submit; Hit@10 of 0.999 on 800 queries; 61 hand-written cases and 4,000 fuzzed sentences run in npm test.

fp16x3 / vLLM | GPU Computing and Inference Performance Sep 2026
- fp16x3 splits operands into fp16 hi/lo pairs with Triton split kernels and compensated GEMMs to keep fp32-level accuracy on tensor cores; 2.875× median speedup across three independent T4 sessions, all 117 accuracy checks passed.
- Traced a vLLM throughput collapse on a GTX 1650 to a cuBLAS fp16 GEMM small-M cliff (99.5 → 4.3 GiB/s from M=1 to 2; 0.83× on a T4 with Tensor Cores); filed issues with PyTorch and vLLM.

EXPERIENCE

Far East Facade | Project Engineer, Engineering Department Jun 2026 - Present
- Built a Python workflow to extract delivery batches and test results from supplier emails, supporting delivery plans for 15 material types.

Caixun Technology | Product Intern Jun 2025 - Aug 2025
- Independently wrote 3 PRDs and prototyped 5 core pages in Axure; proposals were scheduled for development. Tracked fixes and retesting, helping resolve over 10 issues.

SKILLS

Development: Python, TypeScript, React, Fastify/FastAPI, SQL, Git, CI
Agents: tool calling, session scheduling, human approval, tracing/OTLP, policy checks, recovery, RAG and retrieval evaluation
Performance: Triton, CUDA, PyTorch, vLLM, controlled experiments. **Languages:** Chinese (native), English (working)